

A Theory of Economic Slack

Pascal Michailat

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CHAPTER 13.

Slackish business cycle model

Chapter 12 showed that the unemployment gap in US data is generally positive and sharply countercyclical: it widens in recessions and narrows in expansions. The first task of part IV is to explain which aggregate shocks can generate these fluctuations. The second task is to clarify whether and how monetary policy—the main stabilization policy in the United States—can partially or completely undo such shocks so as to stabilize unemployment fluctuations. The final task is to assess whether fiscal policy can support monetary policy in stabilizing unemployment.

This chapter develops the simplest possible slackish business cycle model for that purpose. The model clarifies how aggregate demand and supply determine unemployment, and how monetary policy steers aggregate demand to stabilize unemployment in response to shocks. The model covers both normal times, when monetary policy operates in a conventional way by setting the nominal interest rate, and zero-lower-bound episodes, when monetary policy operates by forward guidance. The model also explains how fiscal policy in the form of public spending is able to stabilize unemployment.

13.1. Overview of the model

The model is organized around one slackish market in which self-employed workers sell their production to other households.

In the model, households produce services and not goods. This simplifies the model a lot because if a service is available and no customer purchases it, the service is just lost

and wasted. Services cannot be stored, so we do not need to worry about inventory of unsold goods and the depreciation of unsold goods. Furthermore, households produce services directly—there are no firms. Hence, we do not need to distinguish between a labor market and a product market: we only have the service market.

One might wonder how realistic it is to assume that households produce only services and not goods. It turns out that it is not such a bad assumption. In US data, there have always been more workers in service-producing industries than in goods-producing industries, but the gap between these two types of industries has widened significantly. From 1943 to today, the number of US workers in goods-producing industries has not changed much: 18.8 million workers in 1943, peaking at 25.0 million workers in 1979, and falling slightly to 21.7 million workers in 2024 (BLS 2025a). By contrast, the number of workers in service-providing industries has increased dramatically, from 23.8 million workers in 1943 to 136.3 million in 2024 (BLS 2025b). So the share of US workers in service-providing industries rose from 56% in 1943, to 72% in 1979, and to 86% in 2024. That is, of all workers, only 14% are producing goods in 2024, whereas 44% were producing goods in 1943. That is why modeling the US economy as a service economy seems like an accurate simplification.

We assume that households cannot produce services for themselves, so households also purchase and consume services produced by other households. The key idea is that a matching function mediates all the trades on the market for services, which produces slack.¹

In the following chapters, we expand this baseline model to refine our understanding of unemployment fluctuations—both their causes and consequences. For instance, in chapter 15, we include separate product and labor markets to trace how slack propagates across markets.

13.2. Slackish market for services

The model is dynamic and organized around a single market for services. The structure of the market is similar to that of the dynamic slackish market model of chapter 8, and of the slackish labor market model of chapter 10.

We assume that there are many large households, each indexed by $j \in [0, 1]$. In each household, $h_j > 0$ workers are part of the labor force: either working or searching for a job. The aggregate number of workers in the labor force is $h = \int_0^1 h_j dj$. The labor force comprises l employed workers and $U = h - l$ unemployed workers.

Each worker in the labor force has the capacity to produce $k > 0$ services per unit time. Here the parameter k describes the productivity of labor. The economy's aggregate capacity is kh services per unit time.

¹This is very much in the spirit of Diamond (1982), who developed a model of the product market in which self-employed workers must trade because they cannot consume their own production.

Services are sold through long-term worker-household relationships. Once workers have matched with a household, they become full-time employees of the household. They remain so until they separate, which occurs at rate $\lambda > 0$. The separations reflect retirements, dismissals, quits, and so on.

To find new employees, each household j posts V_j help-wanted ads. The reason why households must constantly hire help is that they are constantly losing existing workers. Filling a help-wanted ad requires the work of $\kappa > 0$ recruiters per unit time. The recruiters advertise the ads opened by their employer, read applications, and interview and select suitable candidates. The aggregate number of help-wanted ads is $V = \int_0^1 V_j dj$.

From the unemployment level U and vacancy level V , we also define the unemployment rate $u = U/h$ and the vacancy rate $v = V/h$.

The rate at which help-wanted ads and workers match is given by a matching function $m(U, V)$ that satisfies the assumptions laid out in chapter 3. The aggregate tightness is the ratio of the two arguments in the matching function: $\theta = V/U = v/u$.

The aggregate tightness determines the rate at which unemployed workers find jobs, $f(\theta) = m(1, \theta)$, and the rate at which help-wanted ads are filled, $q(\theta) = m(1/\theta, 1)$.

13.3. Equilibrium aggregate supply

The equilibrium aggregate supply is the output level when employment flows are balanced:

$$(13.1) \quad y^s(\theta) = [1 - u(\theta)] kh,$$

where the equilibrium unemployment rate is given by (8.8).

As we saw in chapter 8, the convergence to the equilibrium unemployment rate is rapid. Hence, the actual and equilibrium unemployment rates are indistinguishable: this is why we abstract from the law of motion of the unemployment rate (given by (8.3)) and consider the unemployment rate as a function of tightness (given by (8.8)).

13.4. Inflation rate and interest rates

Before studying aggregate demand, we describe the prices of services and assets in the model.

13.4.1. Inflation rate

At time t , services are sold at a unit price $p(t)$. As in any slackish model, we need to specify a price norm to determine $p(t)$. In a dynamic model, the price norm determines the rate of inflation, $\pi(t) = \dot{p}(t)/p(t)$.

We saw in chapter 5 that in reality market prices respond very sluggishly to changes in market conditions. This appears to be true in the aggregate too. Of course, prices always rise, but the pace at which they do so does not seem to be affected by economic conditions. In figure 1.1, we saw that US inflation barely fell when unemployment spiked during the Great Recession, and then barely rose when unemployment declined. More systematically, US inflation does not seem to respond much to shocks to monetary policy or to fluctuations in unemployment.²

Hence, we simply assume that the price of services grows at a fixed rate. This price norm could well be the result of constant communication from the central bank, which always tries to convince people that they will do anything to keep inflation at a fixed level—2% in the United States. Formally, we assume that prices grow at a constant inflation rate π :

$$p(t) = p(0) \exp(\pi t).$$

In chapter 14, we will move away from fixed inflation and introduce endogenous inflation and a Phillips curve to study how fluctuations in slack might affect inflation. For now, we study the model with fixed inflation.

13.4.2. Nominal interest rate

Next, we need to specify how nominal interest rates are set. We assume that the central bank follows an interest rate peg: they maintain the nominal interest rate $i(t)$ at some specific level i at all times in the absence of shocks. We also assume that nominal interest rates are subject to the zero-lower-bound constraint: $i(t) \geq 0$.

13.4.3. Real interest rate

The key price in our model is the real interest rate, because it is the price of government bonds relative to services:

$$r(t) = i(t) - \pi(t).$$

Because the inflation rate and nominal interest rate are fixed, the real interest rate is also fixed at $r = i - \pi$. Note that the real interest rate must be less than the time discount rate ($r < \delta$) because if not, households would not want to consume anything and aggregate demand would collapse.

²For evidence, see Christiano, Eichenbaum, and Evans (1999, figure 2), Stock and Watson (2010, figure 1), and Stock and Watson (2020, figure 1).

13.5. Preference for wealth

In this chapter and the rest of part IV, we make the unusual assumption that wealth enters the utility function. Thanks to this assumption, the aggregate demand is nondegenerate and the model is much more robust. How can we justify such an assumption?

The standard model assumes that people save to smooth consumption over time, but it has long been recognized that people enjoy accumulating wealth irrespective of future consumption. In fact, people accumulate wealth not just because they want to use it for future consumption, but also because they enjoy the social status and power that comes with wealth.

Describing the European upper class of the early 20th century, Keynes (1919, chapter 2) noted that:

The duty of saving became nine-tenths of virtue and the growth of the cake the object of true religion.... Saving was for old age or for your children; but this was only in theory—the virtue of the cake was that it was never to be consumed, neither by you nor by your children after you.

Fisher (1930, p. 17), who invented the model of consumption and saving that we use today, added that:

A man may include in the benefits of his wealth ... the social standing he thinks it gives him, or political power and influence, or the mere miserly sense of possession, or the satisfaction in the mere process of further accumulation.

A few decades later, describing the process of economic development across countries, and the importance of people's willingness to save, Lewis (1955, pp. 26–27) observed that:

In most communities the attractions of asceticism are small when compared with the attractions of wealth, either as a means to power, or as a mark of superior social status.... Wealth is also desired as a means to power—whether it be the power of bribery, political power, power over employees, or other forms of power.... In every country of the world wealth wins respect and prestige, even though in some there is a time lag.

Keynes and Fisher and Lewis all observed that people enjoy accumulating wealth because it offers social status and power. Today, we have scientific evidence to back this claim: neuroscientists have actually confirmed that wealth itself provides utility, independently of the consumption it can buy. Camerer, Loewenstein, and Prelec (2005, p. 32) explain:

Brain-scans conducted while people win or lose money suggest that money activates similar reward areas as do other 'primary reinforcers' like food and

drugs, which implies that money confers direct utility, rather than simply being valued only for what it can buy.

A survey of 2000 Americans aged 62–75 conducted by the Employee Benefit Research Institute in 2020 and analyzed by Coy (2021) confirms these ideas. The survey first asks respondents what they plan to do over the course of their retirement with the wealth that they hold in their financial accounts. About 15% plan to grow their wealth, 1% plan to spend down none of their wealth, and another 35% plan to spend down a small portion of their wealth. So about 60% of the respondents do not plan to spend down their wealth in retirement.

Asked why they do not plan to consume their wealth in old age, about 35% of respondents say that spending down is not necessary, and another 30% say that not spending makes them feel better. The survey explores that finding further and asks respondents whether they agree with the statement that “saving as much as I can makes me happy and fulfilled.” About 65% of the respondents strongly or somewhat agree with the statement—a strong validation of the wealth-in-the-utility assumption.

Since people do value holding wealth, it is reasonable to assume that it enters the utility function. Among all the reasons why people may value wealth, we focus on social status: we postulate that people enjoy wealth because it provides status in society. We therefore introduce relative wealth into the utility function. The assumption seems plausible. Adam Smith, Ricardo, John Rae, J.S. Mill, Marshall, Veblen, and Frank Knight all believed that people accumulate wealth to attain high social status (Steedman 1981). More recently, a broad literature has documented that people seek to achieve high social status, and that accumulating wealth is a prevalent pathway to do so.³

13.6. Equilibrium aggregate demand

We now turn to the aggregate demand in the model, which we derive by solving the consumption-saving problem of households.

13.6.1. Household’s utility function

To study the aggregate demand side of the model, we introduce the utility function for households. Each household j values two things: services and relative real wealth.

The utility from consumption of services, $c_j(t) \geq 0$, is

$$c_j(t)^{1-\alpha},$$

³See for instance Frank (1985), Cole, Mailath, and Postlewaite (1995), Weiss and Fershtman (1998), Heffetz and Frank (2011), Fiske (2010), Anderson, Hildreth, and Howland (2015), Cheng and Tracy (2013), Ridgeway (2014), Mattan, Kubota, and Cloutier (2017).

where $\alpha \in (0, 1)$ governs the concavity of the utility over services, which determines diminishing marginal utility of consumption.

Each household holds $B_j(t) \geq 0$ government bonds, which is their sole source of wealth. The utility from relative real wealth is simply

$$\frac{1}{a} \cdot \frac{B_j(t) - B(t)}{p(t)},$$

where $B(t) = \int_0^1 B_j(t) dj$ is the average nominal wealth across households, so $B_j(t) - B(t)$ is the household's relative nominal wealth. The relative nominal wealth is divided by the price level $p(t)$ to compute the relative real wealth, $[B_j(t) - B(t)]/p(t)$. The parameter $a > 0$ governs the taste for consuming services relative to holding real wealth. For instance, a higher a means that households value consumption more relative to saving and accumulating wealth and building social status.

Then, the utility that household j maximizes is the discounted sum of flow utilities:

$$(13.2) \quad \int_0^\infty \exp(-\delta t) \left[c_j(t)^{1-\alpha} + \frac{1}{a} \cdot \frac{B_j(t) - B(t)}{p(t)} \right] dt.$$

13.6.2. Household's budget constraint

The next step to deriving the aggregate demand curve is to set up the budget constraint of the household. First, we need to figure out household j 's income. There are two components to income: labor income and saving income. Labor income comes from the $[1 - u(\theta(t))] h_j$ people who are working in the household. These workers sell $k[1 - u(\theta(t))] h_j$ services per unit time. All services have a price $p(t)$, so labor income is

$$p(t) [1 - u(\theta(t))] k h_j.$$

Household j also holds $B_j(t)$ government bonds, which have a nominal interest rate $i(t)$. Hence the household's saving income is

$$i(t) B_j(t).$$

Next, we need to look at the sources of expenditure. First, to consume $c_j(t)$ services, household j must purchase more than $c_j(t)$ services. Indeed, some services are used only for recruiting, to fill help-wanted ads, and do not result in consumption. The amount of services paid for at any time is

$$(13.3) \quad y_j(t) = [1 + \tau(\theta(t))] c_j(t)$$

where $\tau(\theta)$ is the matching wedge, given by (8.13). A household spends $p(t) [1 + \tau(\theta(t))] c_j(t)$ on services at time t : $c_j(t)$ of these services are consumed while $\tau(\theta(t))c_j(t)$ of these services are used for recruiting.

Additionally, to finance the interest payments on government bonds, the government levies a lump-sum tax $T(t)$ on each household.

Bringing together the household's income and expenditure, we build the household's nominal budget constraint:

$$(13.4) \quad \dot{B}_j(t) = iB_j(t) + p(t) [1 - u(\theta(t))] kh_j - p(t) [1 + \tau(\theta(t))] c_j(t) - T(t),$$

where $\dot{B}_j(t)$ is the change in savings, or nominal wealth, at time t .

To express the budget constraint in real terms, we introduce the household's real stock of bonds,

$$b_j(t) = \frac{B_j(t)}{p(t)}.$$

A useful relationship is $\ln(b_j(t)) = \ln(B_j(t)) - \ln(p(t))$. Taking the derivative of this relationship with respect to time, we get

$$\frac{\dot{b}_j(t)}{b_j(t)} = \frac{\dot{B}_j(t)}{B_j(t)} - \frac{\dot{p}(t)}{p(t)} = \frac{\dot{B}_j(t)}{B_j(t)} - \pi.$$

Thus, the temporal change in real wealth is linked to the temporal change in nominal wealth by

$$(13.5) \quad \dot{b}_j(t) = \frac{\dot{B}_j(t)}{p(t)} - \pi b_j(t),$$

because $b_j(t)/B_j(t) = 1/p(t)$.

By combining the nominal budget constraint (13.4) with equation (13.5), we finally obtain the law of motion of real wealth:

$$(13.6) \quad \dot{b}_j(t) = r b_j(t) + [1 - u(\theta(t))] kh_j - [1 + \tau(\theta(t))] c_j(t) - \frac{T(t)}{p(t)}.$$

13.6.3. Household's problem

We are now ready to solve household j 's problem, which will bring us closer to deriving the model's aggregate demand. The household's problem is to choose a path for consumption $c_j(t) \geq 0$ and for real wealth $b_j(t) \geq 0$ to maximize utility (13.2) subject to the budget constraint (13.6). The household takes as given the paths of aggregate variables $\theta(t)$, $r(t)$, $b(t)$, and $T(t)/p(t)$, and the endowment of real wealth $b_j(0) \geq 0$. The household is

also subject to a no-Ponzi condition that prevents them from running Ponzi schemes:
 $\lim_{t \rightarrow \infty} \exp(-rt)b_j(t) \geq 0$.

To solve this optimal control problem, we set up a Hamiltonian (result A.25):

$$\begin{aligned} \mathcal{H}(t, c_j(t), b_j(t)) = & c_j(t)^{1-\alpha} + \frac{1}{a} [b_j(t) - b(t)] \\ & + \psi_j(t) \left\{ rb_j(t) + [1 - u(\theta(t))] kh_j - [1 + \tau(\theta(t))] c_j(t) - \frac{T(t)}{p(t)} \right\}. \end{aligned}$$

The variable $\psi_j(t)$ is the costate variable on household j 's real budget constraint.

We now introduce the first-order conditions that characterize the household's optimal behavior. First, $\partial \mathcal{H} / \partial c_j = 0$, which directly links consumption to the costate variable:

$$(13.7) \quad \psi_j(t) = \frac{1}{1 + \tau(\theta(t))} \cdot (1 - \alpha) c_j(t)^{-\alpha}.$$

From this equation we see the meaning of the costate variable on real wealth, $\psi_j(t)$: it gives the marginal utility derived from one unit of real wealth. Indeed, one unit of real wealth allows the household to purchase one service, which allows them to consume $1/(1 + \tau)$ services. The marginal utility from consumption is $(1 - \alpha) c_j(t)^{-\alpha}$, so the marginal utility from one unit of real wealth is $[1/(1 + \tau)] \times (1 - \alpha) c_j(t)^{-\alpha}$, which is just the costate variable.

The second first-order condition is $\partial \mathcal{H} / \partial b_j = \delta \psi_j(t) - \dot{\psi}_j(t)$, which requires

$$\frac{1}{a} + r \psi_j(t) = \delta \psi_j(t) - \dot{\psi}_j(t).$$

This in turn gives the differential equation describing how the costate variable associated with real wealth evolves when household j saves and spends optimally:

$$(13.8) \quad \dot{\psi}_j(t) = (\delta - r) \psi_j(t) - \frac{1}{a}.$$

In addition to these first-order conditions, optimality requires that the appropriate transversality condition is satisfied: $\lim_{t \rightarrow \infty} \exp(-\delta t) \psi_j(t) b_j(t) = 0$.

Here, any interior solution to the first-order conditions (13.7)–(13.8) is a global maximum of the household's problem. This is because the utility function (13.2) is concave in both consumption and real wealth, and the budget constraint (13.6) is linear in both consumption and real wealth (result A.27).

13.6.4. Euler equation

The differential equation (13.8) implicitly describes household j 's consumption and saving over time: it is the Euler equation of the model. By solving it, we will obtain the model's

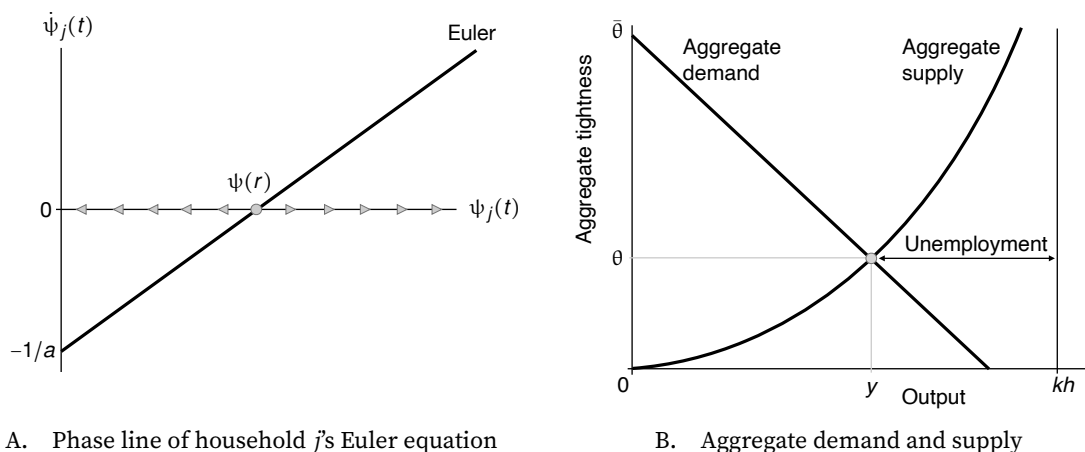


FIGURE 13.1. Solution of the slackish business cycle model

The variable $\psi_j(t)$ is the costate variable associated with the household j 's real wealth. The Euler equation is given by equation (13.8). The equilibrium point $\psi(r)$ is given by equation (13.9). The aggregate supply curve is given by equation (13.1). The aggregate demand curve is given by equation (13.11).

aggregate demand. The Euler equation is an autonomous linear first-order differential equation, and to solve it, we need to understand its dynamical properties. First, its equilibrium point is

$$(13.9) \quad \psi(r) = \frac{1}{(\delta - r)a}.$$

The equilibrium value of the costate variable on real wealth has a simple interpretation. The Euler equation (13.8) shows that people's saving decision are determined by the time discount rate, which says how patient they are, the real interest rate, which determines financial returns on savings, plus marginal utility of real wealth, which determines the hedonic returns on wealth—indeed, with wealth in the utility function, the returns on wealth are not only financial but also hedonic. People keep consumption constant in equilibrium only if they are indifferent between consuming now or later, which requires the hedonic and financial returns on wealth to exactly offset time discounting.

To be precise: instead of consuming one unit of real wealth today, saving it to consume it later add utility $r\psi$ since the real wealth has grown in the future, removes utility $\delta\psi$ since future consumption is discounted, and add utility $1/a$, which holding real wealth today provides some utility directly. These effects must add up to zero in equilibrium: $\delta\psi = r\psi + 1/a$, which is equivalent to (13.9).

Second, since $\delta - r > 0$, the differential equation is a source. These properties appear on the phase line plotted in figure 13.1A.

The variable $\psi_j(t)$ is a costate variable, so it is not a predetermined variable at time t : it can jump at any time. If $\psi_j(t)$ jumps above the equilibrium point, it would start in-

creasing over time and would diverge to ∞ . This means that consumption would decrease and eventually become negative, violating one of the problem's feasibility constraints (see equation (13.7)). If $\psi_j(t)$ jumps below the equilibrium point, it would decrease over time, which means that consumption would increase over time and would diverge to ∞ , eventually violating the resource constraint in the economy. Therefore, $\psi_j(t)$ must jump to the equilibrium point at $t = 0$ —transition to the equilibrium point is immediate. At $t = 0$, $\psi_j(t)$ jumps to $\psi(r)$, so that at all times, $\psi_j(t) = \psi(r)$.

Therefore, using (13.7), we see that at all times t , household j 's consumption is

$$(13.10) \quad c_j(t) = \left[\frac{(1 - \alpha)(\delta - r)a}{1 + \tau(\theta)} \right]^{1/\alpha},$$

where $c_j(t) > 0$. This consumption path is admissible and interior, and it satisfies all the first-order conditions, so it is a global maximum of the household's problem.

13.6.5. Recovering the standard Euler equation

The Euler equation (13.8) looks a bit different from the standard Euler equation in macroeconomic models—which might be worrisome. But there is no need to worry: it is easy to recover the standard form by eliminating the matching wedge and wealth in the utility function.

We remove the matching wedge by setting $\tau(\theta) = 0$ in (13.7) and (13.8), and we remove the utility of wealth by setting $1/a = 0$ in (13.8). The two equations become

$$(1 - \alpha)c(t)^{-\alpha} = \psi(t), \quad \frac{\dot{\psi}(t)}{\psi(t)} = \delta - r(t).$$

Next, we take the logarithm of the first equation and differentiate it with respect to time. We obtain

$$-\alpha \cdot \frac{\dot{c}(t)}{c(t)} = \frac{\dot{\psi}(t)}{\psi(t)}.$$

Combining these equations, we recover the standard Euler equation:

$$\frac{\dot{c}(t)}{c(t)} = \frac{1}{\alpha} \cdot [r(t) - \delta].$$

As usual, the equation says that the growth rate of consumption is determined by the gap between the real interest rate and the time discount rate.

We can also see why the standard Euler equation is degenerate. When $r(t) \neq \delta$, $\dot{c}(t) \neq 0$ for all $c(t) > 0$: the differential equation does not admit an interior equilibrium point. That is, there is no $c(t) > 0$ such that $\dot{c}(t) = 0$. If $r(t) = \delta$, then there is another strange situation: $\dot{c}(t) = 0$ for all $c(t) > 0$. So the household's Euler equation is not well behaved

without wealth in the utility function. This ultimately means that the aggregate demand is degenerate without wealth in the utility function: it imposes that the real interest rate equals the time discount rate ($r = \delta$) but does not determine a consumption level.

13.6.6. Expression of the equilibrium aggregate demand

We have seen that although our model is dynamic, consumption jumps directly to the equilibrium point of the Euler equation at $t = 0$ and stays there. Thus, we can solve the business cycle model by supply-demand analysis, using tightness-quantity diagrams, with a very similar solution method to what we did for the market models.

The first step is to write down the aggregate demand curve, which gives us the aggregate amount of output that is demanded by households according to their Euler equations. The aggregate amount of consumption demanded is $c = \int_0^1 c_j dj$, where the households' demands for consumption c_j are given by (13.10). Then, the aggregate amount of output demanded is $y = [1 + \tau(\theta)]c$. This gives the equilibrium aggregate demand:

$$(13.11) \quad y^d(\theta, i) = \frac{[(1 - \alpha)(\delta + \pi - i)a]^{1/\alpha}}{[1 + \tau(\theta)]^{1/\alpha - 1}}.$$

We see that the aggregate demand is decreasing in aggregate tightness, from $y^d(0, i) > 0$ to $y^d(\bar{\theta}, i) = 0$ as tightness rises from $\theta = 0$ to $\theta = \bar{\theta}$. The aggregate demand is also decreasing in the nominal interest rate i and in fact in the real interest rate $r = i - \pi$.

Note that the aggregate demand is an equilibrium object because the labor flows are balanced at any point in time, so the number of workers who find a job in any household equals the number of workers who lose their job there.

13.7. Solution of the model

We now solve the slackish business cycle model. We distinguish normal times, when the nominal interest rate is positive, and zero-lower-bound episodes, when the nominal interest rate is at zero.

13.7.1. Solution in normal times

The structure of the solution to this chapter's business cycle model is virtually identical to the structure of the solution of the market model of chapter 4. In particular, the key step in solving the model is to find aggregate tightness. Then, from the value of tightness, we can infer the values of all the other variables in the model.

We know that aggregate tightness must equalize aggregate demand and aggregate

supply:

$$(13.12) \quad y^d(\theta, i) = y^s(\theta).$$

The aggregate demand and aggregate supply functions have the same properties as the market demand and market supply in the fixed-price model of chapter 4, and therefore, we can use the same argument to show that equation (13.12) admits a unique solution, and therefore that the business cycle model admits a unique solution.

Accordingly, equation (13.12) implicitly defines tightness as a function of the nominal interest rate, $\theta(i)$. Then, from θ , we can back out all the other variables.

The solution of the model can be represented graphically just as in chapter 4. The tightness that solves the model is at the intersection of the aggregate demand and supply curves, as shown in figure 13.1B. The aggregate supply curve captures the capacity of the economy plus the matching process. The aggregate demand curve captures the fact that households maximize their utility given their budget constraint. At the intersection of the aggregate supply and aggregate demand curves, the matching process is respected (including hiring and separations), and households' utility maximization process is respected.

13.7.2. Solution at the zero lower bound

Our model behaves the same at the zero lower bound and away from it. In both situations, the aggregate demand and supply curves are identical. Indeed, in equations (13.1) and (13.11), nothing changes whether the economy is at the ZLB ($i = 0$) or not ($i > 0$). Hence, the model accommodates permanent zero-lower-bound episodes, and business-cycle shocks and policies have identical effects then, with the obvious difference the conventional monetary policy cannot stimulate the economy further by lowering the nominal interest rate at the zero lower bound.

This property of the slackish business cycle model is entirely different from what occurs in a New Keynesian model. The New Keynesian model predicts that when the zero lower bound lasts sufficiently long, output and inflation collapse to implausibly low levels, and policies such as public spending and forward guidance have extraordinarily large effects (Michaillat and Saez 2021). More generally, in New Keynesian models, the zero lower bound is “a topsy-turvy world, in which many of the usual rules of macroeconomics are stood on their head” (Eggertsson and Krugman 2012, p. 1472). The zero lower bound is not topsy-turvy in slackish models.

13.8. Effects of aggregate shocks

Now that we have characterized the solution in our dynamic model of slack, we use comparative statics to look at the effect of aggregate demand and supply shocks on aggregate

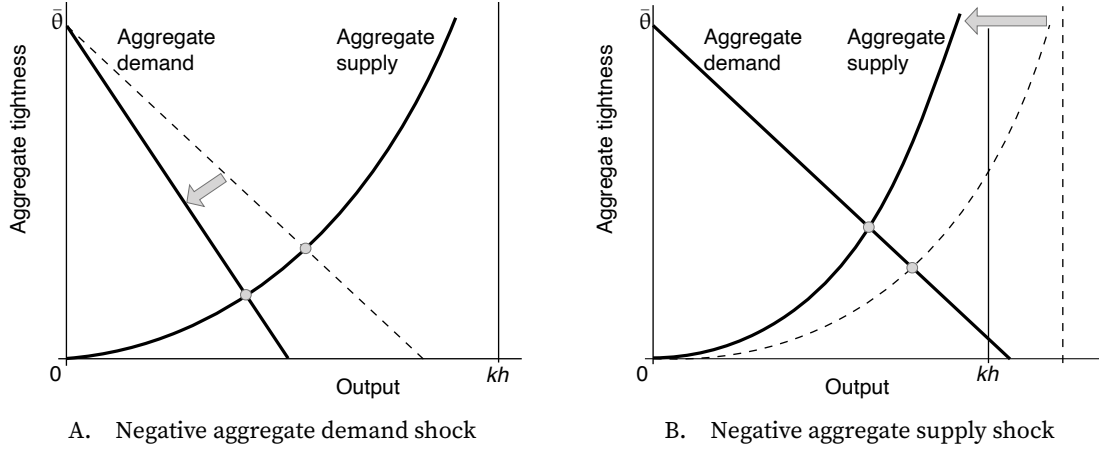


FIGURE 13.2. Aggregate demand and supply shocks in the slackish business cycle model

The aggregate supply curve is given by equation (13.1). The aggregate demand curve is given by equation (13.11).

tightness, output, and employment.

13.8.1. Qualitative effects of aggregate demand shocks

The most natural source of a negative aggregate demand shock is a decrease in the taste for consumption relative to saving, a . Another possible negative aggregate demand shock is a decrease in the time discount rate, δ . After such a shock, households become more thrifty: they desire to save more and consume less, which depresses aggregate demand (equation (13.11)). Graphically, the shock triggers an inward rotation of the aggregate demand curve (figure 13.2A).

We see that as the aggregate demand curve rotates downward, both aggregate tightness θ and output y decrease. The unemployment rate, $u(\theta)$, increases due to a decrease in tightness, as shown by (8.8). A negative demand shock means people want to consume less, so there is less output produced and a lower tightness. There is more slack in the economy, as a larger share of the economy's productive capacity remains unsold and unused.

The efficiency analysis of chapter 9 continues to apply to this model, as the economy is organized around a single market that features a Beveridge curve with two sources of welfare cost: labor services that are unemployed and labor services that are devoted to recruiting instead of consumption. Hence, the efficient aggregate tightness θ^* is given by (9.10): $\theta^* \beta(\theta^*) \kappa = 1$, where $\beta(\theta)$ is the Beveridge elasticity, given by (9.9). The key is that the efficient tightness is not affected by aggregate demand shocks, so the efficient unemployment rate $u^* = u(\theta^*)$ is not affected either. This implies that the unemployment gap $u - u^*$ moves together with the unemployment rate. In particular, the unemployment gap widens when the unemployment rate rises after a negative aggregate demand shock.

TABLE 13.1. Comparative statics in the slackish business cycle model

	Tightness θ	Output y	Employment l	Unemployment u	Gap $u - u^*$
A. Aggregate demand shocks					
Decrease in time discount rate δ	–	–	–	+	+
Decrease in taste for consumption a	–	–	–	+	+
B. Aggregate supply shocks					
Decrease in labor productivity k	+	–	+	–	–
Decrease in labor force h	+	–	–	–	–
C. Policy shocks					
Decrease in nominal interest rate i	+	+	+	–	–
Increase in public spending g	+	+	+	–	–

The comparative statics are obtained from figure 13.2A, figure 13.2B, figure 13.4A, and the adaptation of figure 11.1A to this model. A variable's increase is denoted by “+” and a decrease by “–”.

13.8.2. Quantitative effects of aggregate demand shocks

To quantify the effect of aggregate demand shocks on the unemployment rate, we derive the semielasticity of unemployment with respect to the taste for consumption, ε_a^u . The semielasticity is defined as follows:

$$\varepsilon_a^u = a \cdot \frac{du}{da},$$

and it is related to the elasticity of unemployment with respect to the taste for consumption by $\varepsilon_a^u = u \cdot \varepsilon_a^u$. Semielasticities are typically easier to interpret than elasticities for rate variables that take small values close to zero (such as the unemployment rate or slack rate or interest rate). In terms of notation, we use a curly ε instead of the regular ϵ to denote semielasticities.

Using the analysis of demand shocks in slackish markets, it is straightforward to obtain the elasticity of tightness with respect to the taste for consumption. Formula (6.10) applies here too, so

$$\epsilon_a^\theta = \frac{\epsilon_a^d}{\epsilon_\theta^s - \epsilon_\theta^d},$$

where ϵ_a^d is the elasticity of aggregate demand with respect to the taste for consumption, $\epsilon_a^d = 1/\alpha$, ϵ_θ^d is the elasticity of aggregate demand with respect to tightness, given by (6.13), and ϵ_θ^s is the elasticity of aggregate supply with respect to tightness, given by (8.14). Combining all these elements, we find that the elasticity is

$$\epsilon_a^\theta = \frac{1}{\alpha [1 - \eta(\theta)] u(\theta) + (1 - \alpha) \eta(\theta) \tau(\theta)}.$$

Next, we compute the elasticity of the unemployment rate with respect to tightness from equation (8.8) (using again results from appendix B):

$$(13.13) \quad \epsilon_{\theta}^u = -\frac{f'(\theta)}{f(\theta) + \lambda} [1 - \eta(\theta)] = -[1 - \eta(\theta)] [1 - u(\theta)].$$

Accordingly, the elasticity of the unemployment rate with respect to the taste for consumption is

$$\epsilon_a^u = \epsilon_{\theta}^u \cdot \epsilon_a^{\theta} = -\frac{1 - u(\theta)}{\alpha u(\theta) + (1 - \alpha) \cdot \frac{\eta(\theta)}{1 - \eta(\theta)} \cdot \tau(\theta)}.$$

Last, the semielasticity of the unemployment rate with respect to the taste for consumption is

$$(13.14) \quad \varepsilon_a^u = u \cdot \epsilon_a^u = -\frac{1 - u(\theta)}{\alpha + (1 - \alpha) \cdot \frac{\eta(\theta)}{1 - \eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)}}.$$

The semielasticity ε_a^u gives the percentage-point increase in the unemployment rate when the taste for consumption rises by 1%. We see that it is negative, confirming that the unemployment rate rises when the taste for consumption drops (a negative aggregate demand shock). The effect is sharply dependent on tightness—so the slackish business cycle model is state-dependent.

It's not easy to establish the monotonicity of the semielasticity because both the numerator and denominator are increasing in tightness. But we can obtain a few results, and guess a few properties of the semielasticity too—which we will validate numerically. First, when the economy is operating efficiently, we learn from (9.11) that the semielasticity is just $-(1 - u^*) \approx -1$. Because $u(\theta)$ is small compared to 1, while the two terms in the denominator are commensurate, it is quite likely that the change in the denominator term when θ changes dominates the change in $1 - u(\theta)$ in the numerator, which would then imply that the semielasticity is decreasing in θ around efficiency.

Figure 13.3A validates this conjecture when the slackish business cycle model is calibrated just like the labor market of chapter 10 (table 10.2). Most of the parameters have similar interpretations in the two models, so it seems appropriate to use the calibration. The sole relevant parameter with a different interpretation is α : it determines the shape of the production function in the labor market model and the shape of the utility function in this business cycle model. Both functions are concave, and we keep the same value of α in the two models for transparency and simplicity.

We see that the semielasticity is around -1 when the economy is efficient (tightness around 1). The amplitude of the semielasticity is for the most part decreasing in tightness. The logic is the same as in figure 8.3A: when the economy is slack, changes in aggregate demand have large effects on quantities and thus the unemployment rate; conversely,

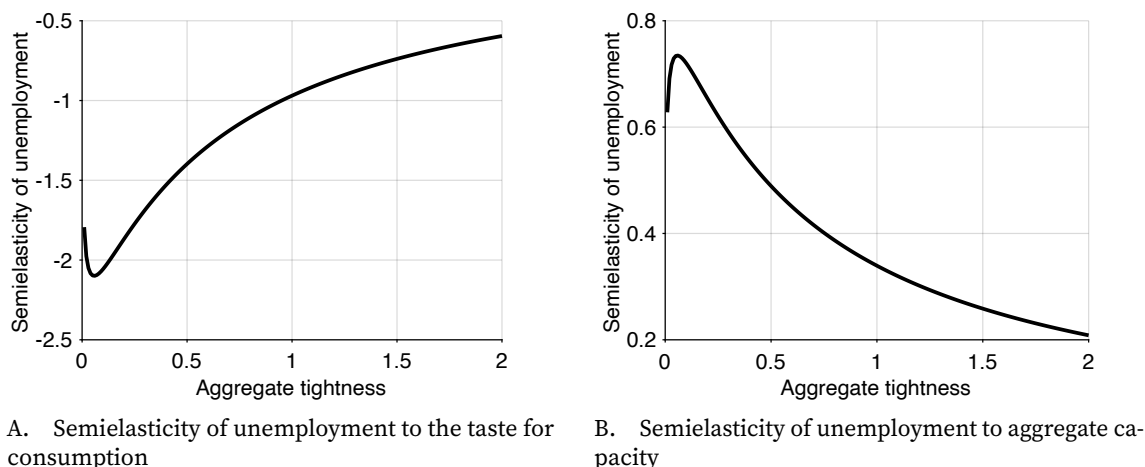


FIGURE 13.3. Effects of aggregate shocks on unemployment in the slackish business cycle model

The semielasticity of unemployment to the taste for consumption is given by (13.14). The semielasticity of unemployment to aggregate capacity is given by (13.15). The model is calibrated as in table 10.2.

when the economy is tight, changes in aggregate demand have small effects on quantities and thus the unemployment rate. The monotonicity breaks down only when tightness is very low and the unemployment rate very high, because then the movements in the numerator of the semielasticity, $1 - u(\theta)$, take over.

13.8.3. Qualitative effects of aggregate supply shocks

The next step is to look at the effects of aggregate supply shocks on the economy and unemployment. A negative aggregate supply shock could be caused by a decrease in either labor productivity, k , or labor force, h .

After a decrease in labor force or productivity, the aggregate capacity shrinks, so the aggregate supply shifts inward (figure 13.2B). As a result, aggregate tightness increases but output decreases. Note that unlike under aggregate demand shocks, tightness and output move in opposite directions for aggregate supply shocks—the usual pattern that we have been observing since chapter 4. The unemployment rate $u(\theta)$ decreases because tightness increases. So here, people want to consume a similar amount of services, but fewer services are offered in the economy, so the economy becomes tighter: there is less slack, and labor services are sold more easily.

Although shocks to productivity and labor force have similar effects on output and tightness, they have a different effect on the employment level. After a reduction in labor force, the number of employed workers l drops: fewer workers want to work, so fewer workers work. Formally: $l = y/k$ and y decreases, so l decreases. After a reduction in productivity, on the other hand, the number of employed workers rises: workers are

less productive so more workers are required to provide demanded services. Formally, $l = (1 - u)h$ and u decreases, so l actually increases here. In this model, the fact that output and employment move in opposite directions is specific to productivity shocks.

13.8.4. Quantitative effects of aggregate supply shocks

To quantify the effect of aggregate supply shocks on the unemployment rate, we derive the semielasticity of unemployment with respect to aggregate capacity, ϵ_{kh}^u . The semielasticity is defined as follows:

$$\epsilon_{kh}^u = kh \cdot \frac{du}{dkh}.$$

It is related to the elasticity of unemployment with respect to aggregate capacity by $\epsilon_{kh}^u = u \cdot \epsilon_{kh}^u$.

Using the analysis of supply shocks in slackish markets, it is straightforward to obtain the elasticity of tightness with respect to aggregate capacity. Formula (6.15) applies here too, so

$$\epsilon_{kh}^\theta = \frac{-\epsilon_{kh}^s}{\epsilon_\theta^s - \epsilon_\theta^d},$$

where ϵ_{kh}^s is the elasticity of aggregate supply with respect to aggregate capacity, $\epsilon_{kh}^s = 1$, ϵ_θ^d is the elasticity of aggregate demand with respect to tightness, given by (6.13), and ϵ_θ^s is the elasticity of aggregate supply with respect to tightness, given by (8.14). (We have also set $\epsilon_{kh}^d = 0$ in (6.15) since aggregate demand does not depend on aggregate capacity.) Combining these elements, we obtain

$$\epsilon_{kh}^\theta = \frac{-1}{[1 - \eta(\theta)] u(\theta) + \frac{1-\alpha}{\alpha} \cdot \eta(\theta) \tau(\theta)}.$$

Finally, using the elasticity of unemployment with respect to tightness given by (13.13), we find that the semielasticity of the unemployment rate with respect to aggregate capacity is

$$(13.15) \quad \epsilon_{kh}^u = u \cdot \epsilon_\theta^u \cdot \epsilon_{kh}^\theta = \frac{1 - u(\theta)}{1 + \frac{1-\alpha}{\alpha} \cdot \frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)}}.$$

The semielasticity ϵ_{kh}^u gives the percentage-point increase in the unemployment rate when the aggregate capacity rises by 1%. We see that it is positive, confirming that the unemployment rate rises when aggregate capacity rises (a positive aggregate supply shock). The effect is sharply dependent on tightness. As with semielasticity (13.14), we see that when the economy is operating efficiently, the semielasticity is just $(1 - u^*)\alpha \approx \alpha$. We also see that the semielasticity is likely decreasing in θ around efficiency.

Figure 13.3B validates these results in the calibrated slackish business cycle model.

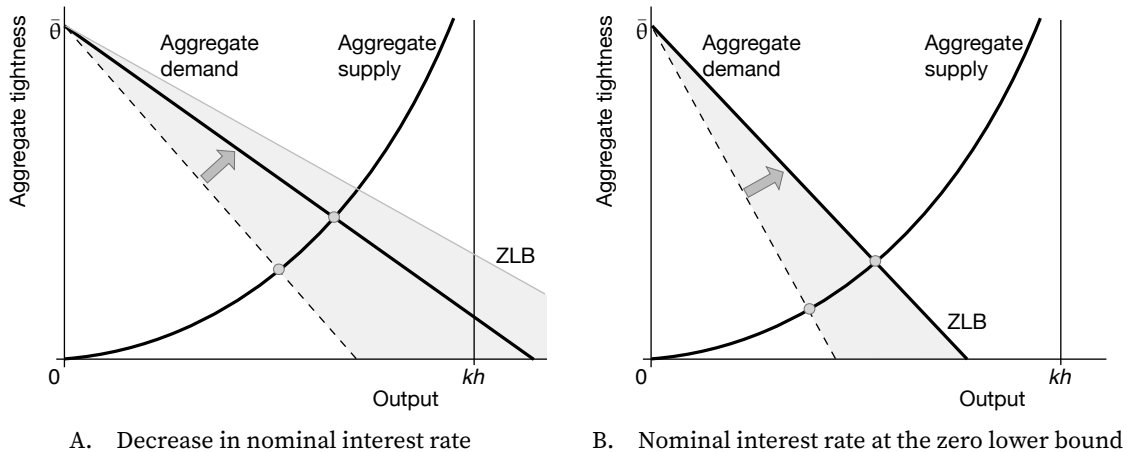


FIGURE 13.4. Expansionary monetary policy in the slackish business cycle model

The aggregate demand and supply curves are constructed in figure 13.1B. The gray cone indicates all the positions that the aggregate demand curve can reach when the nominal interest rate is reduced from its current level. The zero-lower-bound curve shows the position of the aggregate demand curve at the zero lower bound; it is the most outward position that the aggregate demand curve can reach; it is obtained from equation (13.11) when $i = 0$.

The semielasticity is around α when the economy is efficient (tightness around 1). The semielasticity is for the most part decreasing in tightness. Why is that? We know that output responds strongly to supply shocks in tight markets and weakly in slack markets. But the response of output, $y = (1 - u)kh$, is the mechanical response to capacity kh , dampened by the response of the unemployment rate u . So, it follows that there is little dampening in tight markets but significant dampening in slack markets. This then means that the unemployment rate responds little in tight markets but significantly in slack markets—just as the figure shows.

13.9. Monetary policy

We now show how conventional monetary policy steers aggregate demand by modulating the nominal interest rate. In doing so, it affects aggregate tightness and thus the unemployment rate.

13.9.1. How does conventional monetary policy operate?

We start by describing how the model responds to conventional monetary policy. We consider an unexpected permanent decrease in the nominal interest rate i . As inflation π remains constant, the real interest rate $r = i - \pi$ falls. When the real rate falls, wealth has lower returns, so households desire to save less and consume more, which boosts aggregate demand (equation (13.11)). Conventional monetary policy therefore operates

just like an aggregate demand shock.

Graphically, when the nominal interest rate drops, the aggregate demand curve rotates outward, so output and tightness increase (figure 13.4A). Since tightness rises, the unemployment rate decreases. And since the efficient tightness and unemployment rate are unaffected by monetary policy, the unemployment gap moves with the unemployment rate: it shrinks whenever the nominal interest rate is reduced.

13.9.2. The monetary multiplier

Next, we quantify the effect of conventional monetary policy on the unemployment rate. We compute and study the monetary multiplier,

$$\frac{du}{di}.$$

The monetary multiplier gives the percentage-point increase in the unemployment rate when the nominal interest rate increases by 1pp.

We follow the usual steps to compute the monetary multiplier. Formula (6.10) applies here too, so we get the elasticity of tightness with respect to the interest rate:

$$\epsilon_i^\theta = \frac{\epsilon_i^d}{\epsilon_\theta^s - \epsilon_\theta^d},$$

where ϵ_i^d is the elasticity of aggregate demand with respect to the interest rate, given by

$$\epsilon_i^d = -\frac{1}{\alpha} \cdot \frac{i}{\delta + \pi - i},$$

and ϵ_θ^d and ϵ_θ^s are the elasticities of aggregate demand and supply with respect to tightness, given by (6.13) and (8.14). Combining these elements, we find that the elasticity is

$$\epsilon_i^\theta = -\frac{i}{\delta + \pi - i} \cdot \frac{1}{\alpha [1 - \eta(\theta)] u(\theta) + (1 - \alpha) \eta(\theta) \tau(\theta)}.$$

Next, introducing the elasticity of unemployment with respect to tightness, given by (13.13), we find that the monetary multiplier is

$$(13.16) \quad \frac{du}{di} = \frac{u}{i} \cdot \epsilon_\theta^u \cdot \epsilon_i^\theta = \frac{1 - u(\theta)}{\delta - r} \cdot \frac{1}{\alpha + (1 - \alpha) \cdot \frac{\eta(\theta)}{1 - \eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)}}.$$

We see that the monetary multiplier takes the same expression as the semielasticity of unemployment with respect to the taste for consumption, given by (13.14), up to the $-1/(\delta - r)$ factor. This means that the properties of the multiplier follow from those of the

semielasticity. First, the monetary multiplier is positive, confirming that the unemployment rate rises when the nominal interest rate increases. Second, the monetary multiplier is also sharply slack-dependent. When the economy is operating efficiently, the multiplier is $(1 - u^*)/(\delta - r)$, and around efficiency, the multiplier is likely to decrease with tightness.

We plot the monetary multiplier in the calibrated model in figure 13.5. To do that, however, we also need to calibrate the real interest rate r and time discount rate δ . As of 2016, in the United States, the natural rate of interest (real interest rate consistent with full employment) is about $r = 0.5\%$ (Williams 2017, table 1). We use that value for the real interest rate. Estimates of the time discount rate δ obtained in field and laboratory experiments and from real-world behavior are quite dispersed, but the majority of them point to discount rates well above prevailing real interest rates. Frederick, Loewenstein, and O'Donoghue (2002, table 1) and Andersen et al. (2014, table 3) survey a total of 54 studies. For each of the 54 studies we compute the mean estimate, then take the median across studies. We obtain a median annual discount rate of $\delta = 36\%$. The estimates are very dispersed: the lower quartile is $\delta = 13\%$ and the upper quartile is $\delta = 175\%$. To capture the impatience observed in many studies, we also compute a mean discount rate, after excluding 9 studies whose estimates are implausibly high. We obtain a truncated mean of $\delta = 91\%$.⁴ We adopt $\delta = 91\%$ as a baseline calibration.

We see that the monetary multiplier is just above 1 when the economy is efficient (tightness around 1). The amplitude of the monetary multiplier is for the most part decreasing in tightness, from $du/di = 2.3$ when tightness is 0.1 down to $du/di = 0.7$ when tightness is 2. The logic is the same as in figure 8.3A: when the economy is slack, changes in aggregate demand have large effects on quantities and thus the unemployment rate; conversely, when the economy is tight, changes in aggregate demand have small effects on quantities and thus the unemployment rate.

13.10. Forward guidance at the zero lower bound

In this section, we move away from conventional monetary policy and instead study the effects of forward guidance around zero-lower-bound episodes. At the zero lower bound, the nominal interest rate is stuck at zero, so conventional monetary policy does not operate and cannot stimulate the economy. This is why the central bank must turn to forward guidance—to circumvent the zero-lower-bound constraint.

Forward guidance is a type of monetary policy that consists of announcing ahead of

⁴Frederick, Loewenstein, and O'Donoghue (2002, table 1) contains 42 studies and Andersen et al. (2014, table 3) contains 16 studies. There are 4 studies that appear in both tables, so 54 distinct studies are included in total. When we compute the mean discount rate, we remove 9 studies in which respondents are so impatient that the behavior cannot be easily rationalized with an exponential discounting model—they imply immense annual discount rates, sometimes above $10^{10}\%$. The annual discount rate cutoff that we use to remove the 9 studies is 1,000%.

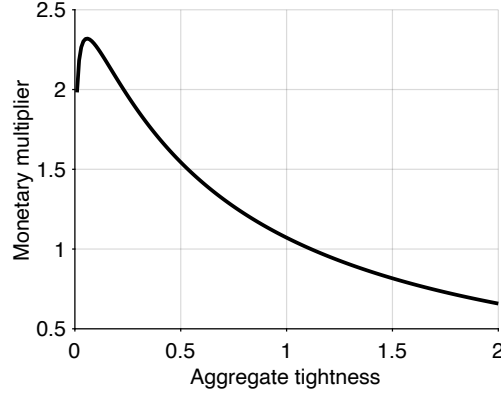


FIGURE 13.5. Monetary multiplier in the slackish business cycle model

The monetary multiplier is given by (13.16). The model is calibrated as in table 10.2, with in addition $\delta = 91\%$ and $r = 0.5\%$.

TABLE 13.2. Timeline of the zero-lower-bound episode and forward guidance

Taste for consumption a	Nominal interest rate i	Equilibrium costate variable ψ
A. Initial normal times: $t < 0$ a_n	$i_n > 0$	$\psi_n = 1/[(\delta + \pi - i_n)a_n]$
B. ZLB episode: $0 < t < t_z$ $a_z < a_n$	0	$\psi_z = 1/[(\delta + \pi)a_z] > \psi_n$
C. Forward guidance: $t_z < t < t_z + \Delta$ a_n	0	$\psi_f = 1/[(\delta + \pi)a_n] < \psi_n$
D. Terminal normal times: $t_z + \Delta < t$ a_n	$i_n > 0$	ψ_n

The parameter $t_z > 0$ gives the duration of the zero-lower-bound episode. The parameter $\Delta > 0$ gives the duration of forward guidance.

time future interest rates. The goal is to make promises that can stimulate the economy today, although the current interest rate is stuck at zero. Forward guidance became popular around the time of the Great Recession because policymakers had no choice—since the zero lower bound became binding in many countries.

13.10.1. Timing of forward guidance

We consider a two-stage scenario where a forward-guidance episode follows a zero-lower-bound episode, as in Cochrane (2017). The complete scenario is presented in table 13.2.

Temporary zero-lower-bound episodes and forward guidance require a dynamic analysis. We therefore come back to the Euler equation (13.8). All households face the same Euler equation, so their costate variables are all the same: $\psi_j(t) = \psi(t)$. The costate

variable $\psi(t)$ indicates how tight the finances of households are, and therefore how much they wish to consume, via equation (13.7).

Importantly, at any point in time, the costate variable determines the aggregate tightness $\theta(t)$. Indeed, from equations (13.7) and (13.3), we see that the costate variable ψ determines the location of the aggregate demand at any point in time:

$$(13.17) \quad y^d(\theta, \psi) = \left[\frac{1 - \alpha}{\psi} \right]^{1/\alpha} \frac{1}{[1 + \tau(\theta)]^{1/\alpha - 1}}.$$

This new expression of the aggregate demand curve is valid for any value of the costate variable, whether it is in equilibrium or not. In fact, when the costate variable is at its equilibrium value $\psi = 1/[(\delta - r)a]$, expression (13.17) reduces to the standard expression for the aggregate demand curve, given by (13.11). Conventional monetary policy moves aggregate demand by changing the interest rate and therefore the equilibrium costate variable; forward guidance moves aggregate demand by changing the costate variable off that equilibrium.

The aggregate demand in forward guidance, (13.17), has the same properties as the aggregate demand curve in normal times, (13.11), except that it depends on the costate variable. The aggregate demand in forward guidance y^d is strictly decreasing in the costate variable ψ .

Of course, aggregate demand must equal aggregate supply at any point in time, and this equality implicitly defines tightness as a function $\theta(\psi)$ of the costate variable:

$$(13.18) \quad y^d(\theta, \psi) = y^s(\theta),$$

where the aggregate demand curve is given by (13.17). The tightness that solves the model is at the intersection of the aggregate demand and supply curves, as shown in figure 13.1B. When the costate variable increases, the aggregate demand curve shifts inward, which lowers tightness along the aggregate supply curve. Conversely, when the costate variable decreases, the aggregate demand curve shifts upward, which raises tightness. Hence, we learn that tightness $\theta(\psi)$ is a strictly decreasing function of the costate variable at any point in time.

Hence, once we determine the dynamics of the costate variable, we can immediately trace the dynamics of aggregate tightness, and then of the unemployment rate and all the other variables in the model. Whenever the costate variable is high, aggregate demand is depressed, so the economy is slack, and conversely, whenever the costate variable is low, the economy is tight.

Before time 0, the economy is in normal times: the taste for consumption is at a high, normal level: $a = a_n$. The nominal interest rate is strictly positive, at $i_n > 0$, to keep the

equilibrium costate variable ψ at a normal level:

$$(13.19) \quad \psi_n = \frac{1}{(\delta + \pi - i_n)a_n}.$$

This keeps the equilibrium tightness at the desired level θ_n , which is related to ψ_n via equation (13.18).

Between times 0 and t_z , there is a zero-lower-bound episode. The taste for consumption is low, depressing aggregate demand: $a = a_z < a_n$. To compensate, the nominal interest rate is lowered to $i = 0$. However, this is insufficient to keep tightness at the desired level: the equilibrium costate variable ψ is elevated, taking a value

$$(13.20) \quad \psi_z = \frac{1}{(\delta + \pi)a_z} > \frac{1}{(\delta + \pi - i_n)a_n} = \psi_n.$$

Accordingly, equilibrium tightness is depressed during the zero-lower-bound episode: $\theta_z < \theta_n$.

To alleviate the situation, the central bank makes a forward-guidance promise at time 0: that it will maintain the nominal interest rate at 0 for a duration Δ once the ZLB is over.

After time t_z , the taste for consumption returns to its normal, higher level: $a = a_n > a_z$. Between times t_z and $t_z + \Delta$, the central bank fulfills its forward-guidance promise and keeps the nominal interest rate at 0. As a result, the equilibrium costate variable ψ is low, taking a value

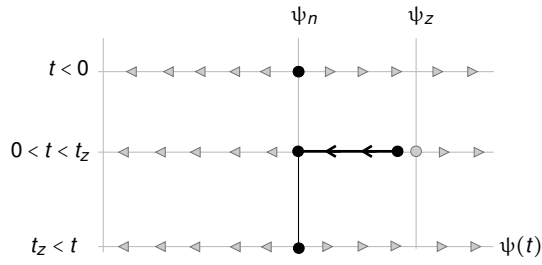
$$(13.21) \quad \psi_f = \frac{1}{(\delta + \pi)a_n} < \frac{1}{(\delta + \pi - i_n)a_n} = \psi_n.$$

After time $t_z + \Delta$, monetary policy returns to normal, $i = i_n$, the equilibrium costate variable is back to its desirable level, ψ_n , and the equilibrium tightness is back to its desirable level, θ_n .

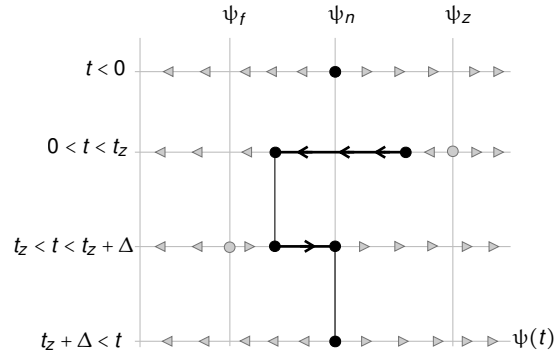
13.10.2. How does the zero lower bound affect the economy?

To fix ideas, we start with a zero-lower-bound episode without forward guidance. We analyze the episode using the phase lines in figure 13.6A. We go backward in time. After time t_z , monetary policy maintains the economy at its desired location by setting the appropriate nominal interest rate. The costate variable $\psi(t)$ is maintained at its normal value ψ_n given by (13.19).

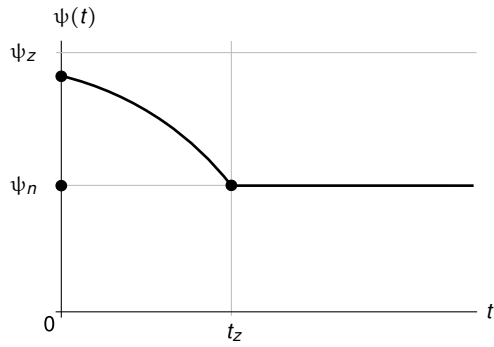
We look next at the zero-lower-bound episode, between times 0 and t_z . Since trajectories are continuous, the economy is at the same point at the end of the zero-lower-bound episode and after it. However, during the zero-lower-bound episode, the point $\psi = \psi_n$ is not an equilibrium: it is below the equilibrium point, so $\dot{\psi} < 0$ at that point. Hence, the



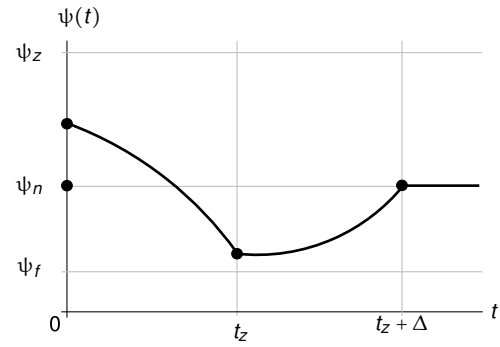
A. Zero lower bound without forward guidance



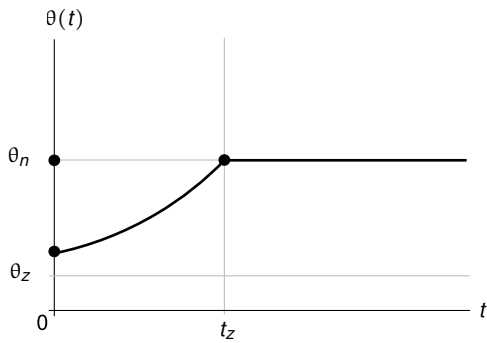
B. Zero lower bound with forward guidance



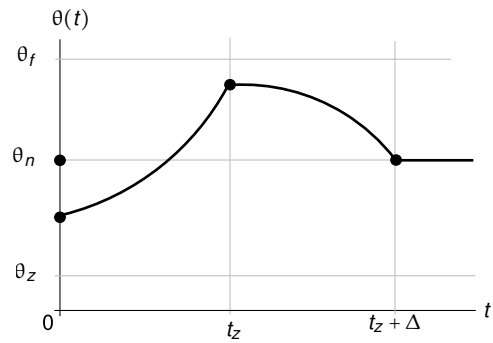
C. Costate variable without forward guidance



D. Costate variable with forward guidance



E. Aggregate tightness without forward guidance



F. Aggregate tightness with forward guidance

FIGURE 13.6. Zero-lower-bound episodes without and with forward guidance

The timeline of the analysis is presented in table 13.2. Panels A and B display the phase lines of the Euler equation (13.8) at the different stages of the analysis. Panels C and D describe how the costate variable $\psi(t)$ is moving over time—as determined from the phase lines. Panels E and F describe how aggregate tightness $\theta(t)$ is moving over time—as inferred from (13.18).

costate variable decreases over time between $t = 0$ and $t = t_z$ to reach that point. From this we infer that at time 0, when the zero-lower-bound episode starts, the costate variable $\psi(0)$ must jump somewhere between the zero-lower-bound equilibrium ψ_z , given by (13.20), and the normal equilibrium ψ_n .

Note that the costate variable can only jump at time 0, because this is the only time when some new information is revealed. Without new information, the optimality of the behavior of households requires that the costate variable evolves continuously over time.

Overall, we see that the costate variable is above normal during the zero-lower-bound episode: it initially jumps to its highest point and then slowly decreases to its normal value (figure 13.6C). Since aggregate tightness is negatively related to the costate variable, as (13.18) shows, we infer that tightness jumps down at time 0, when the taste for consumption jumps down and the zero lower bound becomes binding. After that, the aggregate tightness slowly recovers to reach its normal level at time t_z (figure 13.6E).

13.10.3. How does forward guidance operate?

Now, how can forward guidance improve things at the zero lower bound? Conventional monetary policy improves things by reducing the nominal interest rate to 0 at time 0, as soon as the negative aggregate demand shock hits. But we assume that the decrease in the taste for consumption is so large that the reduction in interest rate is insufficient to maintain the economy at the desired level: $\psi_n < \psi_z$. To supplement conventional monetary policy, the central bank therefore introduces forward guidance.

We analyze the zero-lower-bound episode with forward guidance using the phase lines in figure 13.6B. We go backward in time. After time $t_z + \Delta$, monetary policy maintains the economy at its desired location by setting the appropriate nominal interest rate. The costate variable is maintained at the normal equilibrium ψ_n .

Between times t_z and $t_z + \Delta$, the economy is in forward guidance. Since trajectories are continuous, the economy is at the same point at the end of forward guidance and after it. However, during the forward-guidance episode, the point $\psi = \psi_n$ is not an equilibrium: it is above the equilibrium point, $\psi = \psi_f$, so $\dot{\psi} > 0$ at that point. Hence, the costate variable increases over time between $t = t_z$ and $t = t_z + \Delta$ to reach that point. This means that at the beginning of forward guidance, the costate variable must be below the normal equilibrium ψ_n . We also see that when the forward guidance lasts longer (higher Δ), the costate variable is pushed to lower levels at the beginning of forward guidance.

We then turn to the zero-lower-bound episode, between times 0 and t_z . At time t_z , the costate variable is lower than the normal equilibrium ψ_n and lower than the zero-lower-bound equilibrium ψ_z . This means that $\dot{\psi} < 0$ at that point. Hence, the costate variable decreases over time between $t = 0$ and $t = t_z$ to reach that point. This implies that at time 0, when the zero-lower-bound episode starts, the costate variable $\psi(0)$ must jump

somewhere between the zero-lower-bound equilibrium ψ_z and the forward-guidance equilibrium ψ_f , which is given by (13.21). We also see that when the zero lower bound lasts longer (higher t_z), the costate variable is pushed to higher levels at time 0.

In sum, we see that the costate variable is below normal during the forward-guidance episode. During the zero lower bound, the costate is higher than during forward guidance, but it may be above or below normal, depending on the durations of the zero-lower-bound and forward-guidance episodes. If the zero-lower-bound episode is long enough, the costate variable initially jumps to its highest point, then slowly decreases, and finally increases back to its normal level (figure 13.6D). Since aggregate tightness is negatively related to the costate variable, tightness jumps down at time 0, when the zero lower bound becomes binding. After that, the aggregate tightness slowly increases until time t_z , before decreasing back down to reach its normal level at time $t_z + \Delta$ (figure 13.6F).

The boom engineered during forward guidance therefore improves the situation at the zero lower bound. Instead of reaching the normal equilibrium at time t_z , the economy reaches a point with above-normal tightness at that time, so at any time before t_z , tightness is higher than without forward guidance, so the economy is less slack. Forward guidance is more powerful if it lasts longer and if the zero lower bound is shorter.

13.10.4. Closed-form expressions for the costate variable

So far we have used the phase lines for a qualitative analysis of the zero lower bound and forward guidance. But, because the Euler equation (13.8) is an autonomous linear first-order differential equation, it is not difficult to obtain closed-form expressions for the costate variable. We need to find an expression for the zero-lower-bound episode and a separate expression for the forward-guidance episode. We go backward in time again. From these expressions, we derive the dynamics of the aggregate tightness $\theta(t)$ using equation (13.18).

During forward guidance, the costate variable's dynamics are governed by

$$\dot{\psi}(t) - (\delta + \pi)\psi(t) = -\frac{1}{a_n},$$

with boundary condition $\psi(t_z + \Delta) = \psi_n$. Applying result A.29, we find that during forward guidance, when $t_z < t < t_z + \Delta$, the costate variable is given by

$$\psi(t) = \psi_f + (\psi_n - \psi_f) \exp(-(\delta + \pi)(t_z + \Delta - t)).$$

This expression actually tells us where the costate must be at the end of the zero-lower-bound episode, and at the start of forward guidance. When $t = t_z$, the costate variable

equals

$$(13.22) \quad \psi(t_z) = \psi_f \cdot [1 - \exp(-(\delta + \pi)\Delta)] + \psi_n \cdot \exp(-(\delta + \pi)\Delta).$$

So the costate variable is between the forward-guidance and normal equilibrium points, ψ_f and ψ_n , and it is closer and closer to the forward-guidance equilibrium point, so lower and lower, as the forward guidance lasts longer (higher Δ).

During the zero lower bound, the costate variable's dynamics are governed by

$$\dot{\psi}(t) - (\delta + \pi)\psi(t) = -\frac{1}{a_z},$$

with boundary condition that $\psi(t_z)$ is given by (13.22). Thus, we find that during the zero lower bound, when $0 < t < t_z$, the costate variable is given by

$$(13.23) \quad \psi(t) = \psi_z \cdot [1 - \exp(-(\delta + \pi)(t_z - t))] + \left(\psi_f [1 - \exp(-(\delta + \pi)\Delta)] + \psi_n \exp(-(\delta + \pi)\Delta) \right) \exp(-(\delta + \pi)(t_z - t)).$$

Here we see again that the costate variable is higher when the zero lower bound lasts longer (higher t_z) but lower when forward guidance lasts longer (higher Δ).

13.10.5. Forward-guidance puzzle

One possible reason why forward guidance became popular at the time of the Great Recession is that the New Keynesian model predicts that the policy might be extremely powerful.

Michaillat and Saez (2021, proposition 4) isolates two results that describe the potency of forward guidance in the New Keynesian model. Just as described in table 13.2, consider a zero-lower-bound episode of duration t_z followed by a forward-guidance episode of duration Δ . The first result is that there exists a threshold Δ^* such that a forward-guidance episode longer than Δ^* transforms a zero-lower-bound episode of any duration into a boom. In addition, when forward guidance is longer than Δ^* , a long-enough forward-guidance episode or a long-enough zero-lower-bound episode generates an arbitrarily large boom.

However, in the data, future monetary policy has much more subdued effects on output and inflation (Del Negro, Giannoni, and Patterson 2023). The effects of future monetary policy on current output and inflation are implausibly strong in the New Keynesian model. The gap between the New Keynesian model's prediction and the empirical evidence is known as the forward-guidance puzzle.

The slackish model does not suffer from the forward-guidance puzzle, as forward guidance only has limited effects in the model—especially if the zero-lower-bound episode

lasts a long time. We can use the closed-form expressions above to establish these results.

First, there exists a threshold t_z^* such that a zero-lower-bound episode longer than t_z^* prompts a slump—a costate variable above normal and thus tightness below normal—irrespective of the duration of forward guidance. That is, for $t_z > t_z^*$, and for any Δ , $\psi(0) > \psi_n$, which implies $\theta(0) < \theta_n$ through (13.18).

This can be seen using the expression (13.23) for the costate variable during the zero-lower-bound episode. At time 0, the costate variable jumps to

$$\psi(0) = \psi_z \cdot [1 - \exp(-(\delta + \pi)t_z)] + \psi(t_z) \exp(-(\delta + \pi)t_z),$$

where $\psi(t_z)$ is given by (13.22). The value $\psi(t_z)$ is a weighted average of ψ_f and ψ_n , so the lowest value that it can take, irrespective of the duration of forward guidance, is ψ_f (which is reached if the forward-guidance episode lasts an infinitely long time). Since $\psi(0)$ is itself a weighted average of $\psi(t_z)$ and ψ_n , a lower bound on $\psi(0)$ is

$$(13.24) \quad \psi_z \cdot [1 - \exp(-(\delta + \pi)t_z)] + \psi_f \exp(-(\delta + \pi)t_z).$$

That lower bound is a weighted average of ψ_f and ψ_z , and the weight on $\psi_z > \psi_f$ is larger when t_z is larger. Thus, when t_z is large enough, $\psi(0)$ is close enough to ψ_z and therefore above ψ_n . In fact the threshold t_z^* is such that

$$\psi_z \cdot [1 - \exp(-(\delta + \pi)t_z^*)] + \psi_f \exp(-(\delta + \pi)t_z^*) = \psi_n,$$

guaranteeing that any zero-lower-bound episode longer than t_z^* prompts an initial slump. Reshuffling the equation, we extract a closed-form expression for the threshold:

$$t_z^* = \frac{1}{\delta + \pi} \cdot \ln \left(\frac{\psi_z - \psi_f}{\psi_z - \psi_n} \right).$$

Any zero-lower-bound episode longer than t_z^* generates a slump, irrespective of the length of the ensuing forward-guidance episode.

Furthermore, the slump approaches the zero-lower-bound equilibrium as the zero-lower-bound duration approaches infinity, irrespective of forward guidance. We see this by looking at the lower bound on the costate variable given by (13.24). As $t_z \rightarrow \infty$, the lower bound converges to ψ_z , which means that $\psi(0)$ itself converges to ψ_z —as $\psi(0)$ is sandwiched between the lower bound and ψ_z . So when the zero lower bound lasts a very long time, there is nothing that forward guidance can do: it is entirely ineffective, and the economy is stuck in the vicinity of the zero-lower-bound equilibrium.

Finally, arbitrarily large booms are impossible in the slackish model. The costate variable always remains between ψ_f and ψ_z , as (13.23) shows, so the costate variable is

bounded below by ψ_f , implying that tightness is bounded above by θ_f .

13.11. Fiscal policy

Monetary policy plays a central role for stabilization in the United States and most other developed economies, but it is not always sufficient. In particular, when the nominal interest rate is constrained at the zero lower bound and aggregate demand remains too weak, it is natural to ask whether fiscal policy can provide additional support. Such support is especially important if the zero lower bound is expected to last a long time—because then the effectiveness of forward guidance diminishes greatly.

Indeed, the fiscal mechanism developed in chapter 11 in the context of public employment carries over directly to public spending. To simplify, we assume that public spending g does not affect the demand for private goods and services $y^d(\theta, i)$. This is what happens for instance if the utility over public goods and services g and private goods and services c is separable:

$$(1 - \varkappa) \cdot c^{1-\alpha} + \varkappa \cdot g^{1-\alpha},$$

where $\varkappa \in (0, 1)$ measures the value of public goods relative to private goods.

Then, if the government purchases an amount g of goods and services to stimulate the economy, aggregate demand is boosted and becomes

$$(13.25) \quad y^d(\theta, i) + g,$$

where $y^d(\theta, i)$ is the private demand given by (13.11). As public spending stimulates aggregate demand, it also stimulates aggregate tightness, given by the modified equation:

$$(13.26) \quad y^d(\theta, i) + g = y^s(\theta).$$

As tightness rises, the unemployment rate, still given by (8.8), falls.

Hence, in this business cycle model, public spending operates just like public employment in the labor market model of chapter 11. The effect of public spending can be illustrated in the tightness-output diagram. As shown in figure 13.7A, after an increase in public spending, the aggregate demand curve shifts outward and the economy moves up along the aggregate supply curve: aggregate tightness and output rise. As tightness rises, unemployment falls. However, as tightness rises, it becomes more difficult and expensive for households to procure services: households therefore reduce their purchases. Just like public employment crowds out private employment, public spending crowds out private spending (a movement up along the private demand curve). The effects of public spending are summarized in table 13.1.

In fact the multiplier computed in chapter 11 continues to apply in this setting. It

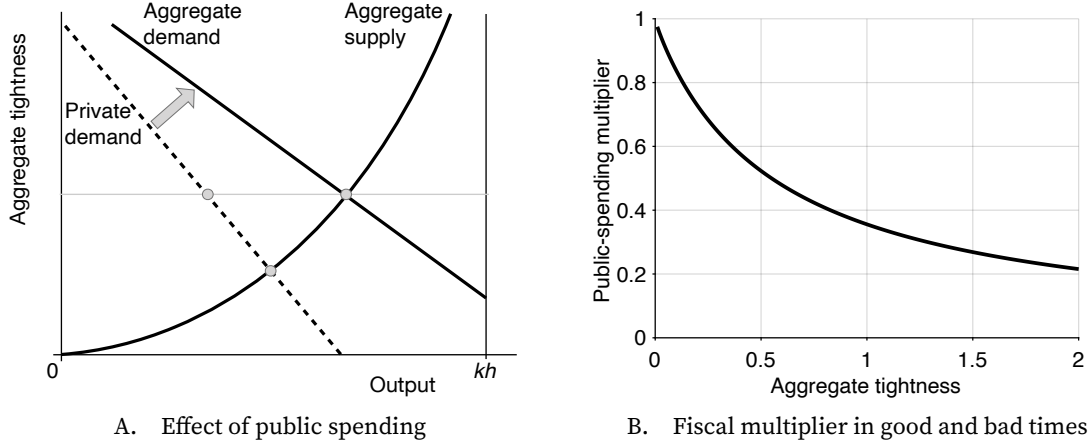


FIGURE 13.7. Fiscal policy in the slackish business cycle model

The aggregate supply curve is given by equation (13.1). The aggregate demand curve is given by equation (13.11). The fiscal multiplier is given by (13.27). The model is calibrated as in table 10.2, with in addition $\delta = 91\%$ and $r = 0.5\%$.

remains the case, as in (11.2), that the public-spending multiplier is given by

$$(13.27) \quad \frac{dy}{dg} = \frac{1}{1 - \epsilon_{\theta}^d / \epsilon_{\theta}^s},$$

where ϵ_{θ}^d is the elasticity of private aggregate demand with respect to aggregate tightness, still given by (6.13), and ϵ_{θ}^s is the elasticity of aggregate supply with respect to aggregate tightness, still given by (8.14). Hence, the properties of the public-spending multiplier are determined by the positive elasticity ratio

$$-\frac{\epsilon_{\theta}^d}{\epsilon_{\theta}^s} = \frac{1 - \alpha}{\alpha} \cdot \frac{\eta(\theta)}{1 - \eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)}.$$

In chapter 8, we showed that the elasticity ratio determines the effects of demand and supply shocks in the slackish market model. Here, it determines all the properties of the public-spending multiplier.

Just like the public-employment multiplier, the public-spending multiplier is strictly positive, strictly less than 1, and strictly decreasing with tightness. The multiplier is α when tightness is efficient. The numerical values of the public-spending multiplier under the same calibration as above are presented in figure 13.7B. The fiscal multiplier is close to 1 when tightness is close to 0, it drops to 0.84 when tightness is 0.1, and then to 0.52 when tightness is 0.5. When the economy is efficient (tightness around 1), the multiplier is 0.35. The multiplier shrinks further as the economy becomes tighter: down to 0.27 when tightness is 1.5 and to 0.22 when tightness is 2.

Hence, purchasing public services increases output and reduces unemployment, but it diminishes purchases of private services by households. There is some crowding out of private services by public services: when the government purchases public services, it makes it harder for households to purchase private services because the government and households compete to hire the same workers.

Furthermore, when tightness is lower, public spending crowds out private spending less and reduces unemployment more. So the public-spending multiplier is slack dependent, as public spending stimulates output more effectively in a slack economy than in a tight economy. This result is supported by several US studies, including Auerbach and Gorodnichenko (2012), Candelon and Lieb (2013), Fazzari, Morley, and Panovska (2015), and Ghassibe and Zanetti (2022). Relatedly, Brunet (2026) finds that employment multipliers were much lower during World War 2 than in later decades in the United States. This is consistent with our model, as the US labor market was the tightest in history during World War 2 (figure 12.1).

13.12. Summary

This chapter develops a slackish model of a service-based economy in which households both produce and consume services. Because all transactions must pass through a matching function, households cannot sell all of their productive capacity, giving rise to slack. The assumption that households derive utility not only from consumption but also from wealth ensures that aggregate demand is well defined. Aggregate tightness equilibrates aggregate demand and aggregate supply, and it in turn determines output, consumption, and unemployment.

The model delivers transparent comparative statics. Negative demand shocks reduce tightness and output, and raise unemployment. Negative supply shocks have distinct signatures: declines in labor force participation and declines in productivity both reduce output and raise tightness, but they have opposite implications for employment: lower productivity raises employment while lower participation decreases it. Finally, the chapter shows that the model continues to behave perfectly normally at the zero lower bound—although of course conventional monetary policy becomes ineffective then.

At the zero lower bound, forward guidance can stimulate the economy, especially when the zero-lower-bound episode is short. The model dynamics during forward guidance are determined by the Euler equation for the costate variable associated with households' wealth. The costate variable shifts the aggregate demand curve and therefore determines aggregate tightness at each date through the aggregate-demand-equals-aggregate-supply equation. This mapping allows us to study temporary zero-lower-bound episodes and forward guidance using transition paths for tightness.

Fiscal policy also stabilizes the economy in the model. Public spending stimulates

aggregate demand, raising tightness and reducing unemployment. The public-spending multiplier—which measures the increase in output for each service purchased by the government—is positive, below one, and larger when the economy is slack.

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